

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 3

### 1. Identification

**Product Name** (S)-(+)-1-Boc-3-fluoropyrrolidine

**Cat No. :** H66889

**CAS-No** 479253-00-4  
**Synonyms** No information available

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |            |
|------------------------------------------|------------|
| <b>Acute oral toxicity</b>               | Category 3 |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 2 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Toxic if swallowed  
Causes serious eye irritation

**Precautionary Statements****Prevention**

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear eye/face protection

**Response**

IF SWALLOWED: Immediately call a POISON CENTER/doctor

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Rinse mouth

If eye irritation persists: Get medical advice/attention

**Storage**

Store locked up

**Disposal**

Dispose of contents/container to an approved waste disposal plant

**Other Hazards**

Very toxic to aquatic organisms

### 3. Composition/Information on Ingredients

| Component                         | CAS-No      | Weight % |
|-----------------------------------|-------------|----------|
| (S)-(+)-1-Boc-3-fluoropyrrolidine | 479253-00-4 | <=100    |

### 4. First-aid measures

**General Advice**

Show this safety data sheet to the doctor in attendance. Immediate medical attention is required.

**Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. In the case of contact with eyes, rinse immediately with plenty of water and seek medical advice.

**Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. Immediate medical attention is required.

**Inhalation**

Remove to fresh air. If not breathing, give artificial respiration. Do not use mouth-to-mouth method if victim ingested or inhaled the substance; give artificial respiration with the aid of a pocket mask equipped with a one-way valve or other proper respiratory medical device. Immediate medical attention is required.

**Ingestion**

Do NOT induce vomiting. Call a physician or poison control center immediately.

**Most important symptoms/effects  
Notes to Physician**

None reasonably foreseeable.  
Treat symptomatically

### 5. Fire-fighting measures

**Unsuitable Extinguishing Media**

No information available

|                                  |                          |
|----------------------------------|--------------------------|
| Flash Point                      | 103 °C / 217.4 °F        |
| Method -                         | No information available |
| Autoignition Temperature         | No information available |
| Explosion Limits                 |                          |
| Upper                            | No data available        |
| Lower                            | No data available        |
| Sensitivity to Mechanical Impact | No information available |
| Sensitivity to Static Discharge  | No information available |

**Specific Hazards Arising from the Chemical**

Do not allow run-off from fire-fighting to enter drains or water courses.

**Hazardous Combustion Products**

Carbon monoxide (CO). Carbon dioxide (CO<sub>2</sub>). Nitrogen oxides (NO<sub>x</sub>). Hydrogen fluoride.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear. Thermal decomposition can lead to release of irritating gases and vapors.

**NFPA**

Health  
3

Flammability  
1

Instability  
0

Physical hazards  
-

## 6. Accidental release measures

|                                  |                                                                                                                                                                                                                                             |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>      | Ensure adequate ventilation. Use personal protective equipment as required. Keep people away from and upwind of spill/leak. Evacuate personnel to safe areas.                                                                               |
| <b>Environmental Precautions</b> | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |

**Methods for Containment and Clean Up** Soak up with inert absorbent material. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

|                 |                                                                                                                                                                                                                                             |
|-----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Use only under a chemical fume hood. Do not breathe mist/vapors/spray. Do not ingest. If swallowed then seek immediate medical assistance. |
| <b>Storage.</b> | Keep containers tightly closed in a dry, cool and well-ventilated place.                                                                                                                                                                    |

## 8. Exposure controls / personal protection

|                                      |                                                                                                                                                                                                                                                                                                                                                                                             |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Exposure Guidelines</b>           | This product does not contain any hazardous materials with occupational exposure limit established by the region specific regulatory bodies.                                                                                                                                                                                                                                                |
| <b>Engineering Measures</b>          | Ensure that eyewash stations and safety showers are close to the workstation location. Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source |
| <b>Personal protective equipment</b> |                                                                                                                                                                                                                                                                                                                                                                                             |
| <b>Eye Protection</b>                | Goggles                                                                                                                                                                                                                                                                                                                                                                                     |
| <b>Hand Protection</b>               | Wear appropriate protective gloves and clothing to prevent skin exposure.                                                                                                                                                                                                                                                                                                                   |

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Nitrile rubber | See manufacturers | -               | Splash protection only |
| Neoprene       | recommendations   |                 |                        |
| Natural rubber |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

#### Respiratory Protection

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Organic gases and vapours filter Type A Brown conforming to EN14387

When RPE is used a face piece Fit Test should be conducted

#### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

#### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                                     |
|----------------------------------------|-------------------------------------|
| Physical State                         | Liquid                              |
| Appearance                             | Colorless - Light yellow            |
| Odor                                   | No information available            |
| Odor Threshold                         | No information available            |
| pH                                     | No information available            |
| Melting Point/Range                    | No data available                   |
| Boiling Point/Range                    | 74 - 80 °C / 165.2 - 176 °F @ 2mmHg |
| Flash Point                            | 103 °C / 217.4 °F                   |
| Evaporation Rate                       | No information available            |
| Flammability (solid,gas)               | Not applicable                      |
| Flammability or explosive limits       |                                     |
| Upper                                  | No data available                   |
| Lower                                  | No data available                   |
| Vapor Pressure                         | No information available            |
| Vapor Density                          | No information available            |
| Specific Gravity                       | 1.069 g/cm3                         |
| Solubility                             | No information available            |
| Partition coefficient; n-octanol/water | No data available                   |
| Autoignition Temperature               | No information available            |
| Decomposition Temperature              | No information available            |
| Viscosity                              | No information available            |
| Molecular Formula                      | C9 H16 F N O2                       |
| Molecular Weight                       | 189.23                              |

## 10. Stability and reactivity

|                 |                                            |
|-----------------|--------------------------------------------|
| Reactive Hazard | None known, based on information available |
| Stability       | Moisture sensitive.                        |

|                                  |                                                                                                                |
|----------------------------------|----------------------------------------------------------------------------------------------------------------|
| Conditions to Avoid              | Incompatible products.                                                                                         |
| Incompatible Materials           | Strong oxidizing agents                                                                                        |
| Hazardous Decomposition Products | Carbon monoxide (CO), Carbon dioxide (CO <sub>2</sub> ), Nitrogen oxides (NO <sub>x</sub> ), Hydrogen fluoride |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                                                                       |
| Hazardous Reactions              | None under normal processing.                                                                                  |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

#### Component Information

Toxicologically Synergistic Products No information available

#### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                 |                                                                                          |
|-----------------|------------------------------------------------------------------------------------------|
| Irritation      | No information available                                                                 |
| Sensitization   | No information available                                                                 |
| Carcinogenicity | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                         | CAS-No      | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|-----------------------------------|-------------|------------|------------|------------|------------|------------|
| (S)-(+)-1-Boc-3-fluoropyrrolidine | 479253-00-4 | Not listed | Not listed | Not listed | Not listed | Not listed |

|                                            |                                                                |
|--------------------------------------------|----------------------------------------------------------------|
| Mutagenic Effects                          | No information available                                       |
| Reproductive Effects                       | No information available.                                      |
| Developmental Effects                      | No information available.                                      |
| Teratogenicity                             | No information available.                                      |
| STOT - single exposure                     | None known                                                     |
| STOT - repeated exposure                   | None known                                                     |
| Aspiration hazard                          | No information available                                       |
| Symptoms / effects, both acute and delayed | No information available                                       |
| Endocrine Disruptor Information            | No information available                                       |
| Other Adverse Effects                      | The toxicological properties have not been fully investigated. |

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms. The product contains following substances which are hazardous for the environment.

|                               |                           |
|-------------------------------|---------------------------|
| Persistence and Degradability | No information available  |
| Bioaccumulation/ Accumulation | No information available. |
| Mobility                      | No information available. |

### 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

### 14. Transport information

#### DOT

UN-No UN2810  
 Proper Shipping Name Toxic liquid, organic, n.o.s.  
 Technical Name ((S)-(+)-1-Boc-3-fluoropyrrolidine)  
 Hazard Class 6.1  
 Packing Group III

#### TDG

UN-No UN2810  
 Proper Shipping Name Toxic liquid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

#### IATA

UN-No UN2810  
 Proper Shipping Name TOXIC LIQUID, ORGANIC, N.O.S.\*  
 Hazard Class 6.1  
 Packing Group III

#### IMDG/IMO

UN-No UN2810  
 Proper Shipping Name Toxic liquid, organic, n.o.s.  
 Hazard Class 6.1  
 Packing Group III

### 15. Regulatory information

#### International Inventories

| Component                         | CAS-No      | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS | ELINCS | NLP |
|-----------------------------------|-------------|-----|------|------|-----------------------------------------------|--------|--------|-----|
| (S)-(+)-1-Boc-3-fluoropyrrolidine | 479253-00-4 | -   | -    | -    | -                                             | -      | -      | -   |

| Component                         | CAS-No      | IECSC | KECL | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|-----------------------------------|-------------|-------|------|------|------|------|------|-------|-------|
| (S)-(+)-1-Boc-3-fluoropyrrolidine | 479253-00-4 | -     | -    | -    | -    | -    | -    | -     | -     |

#### Legend:

X - Listed '-' - Not Listed

**KECL** - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

**DSL/NDSL** - Canadian Domestic Substances List/Non-Domestic Substances List

**TSCA** - United States Toxic Substances Control Act Section 8(b) Inventory

**EINECS/ELINCS** - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

**IECSC** - Chinese Inventory of Existing Chemical Substances

**KECL** - Korean Existing and Evaluated Chemical Substances

**ENCS** - Japanese Existing and New Chemical Substances

**AICS** - Australian Inventory of Chemical Substances

**PICCS** - Philippines Inventory of Chemicals and Chemical Substances

#### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

#### Other International Regulations

**Authorisation/Restrictions according to EU REACH**

Not applicable

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                         | CAS-No      | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|-----------------------------------|-------------|----------------|------------------------------|---------------------------|--------------------------------------------|
| (S)-(+)-1-Boc-3-fluoropyrrolidine | 479253-00-4 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component                         | CAS-No      | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|-----------------------------------|-------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| (S)-(+)-1-Boc-3-fluoropyrrolidine | 479253-00-4 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

## Prepared By

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
www.thermofisher.com

## Revision Date

01-April-2024

## Print Date

01-April-2024

## Revision Summary

New emergency telephone response service provider.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**